

# Beyond Random Sampling: Improving U-Net Performance on Noisy Plant Root Images

Individual Research Project — Block C Year 2 (Second opportunity)

AUTHOR

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GROUP

YEAR

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## INTRODUCTION

Measuring plant root morphology manually is extremely time-consuming. U-Net automates root segmentation, but standard random patch sampling creates a critical problem:

- 85.5% of random patches contain zero root pixels
- Model trains mostly on uninformative background
- Mean root pixel ratio is only 0.71%

No study has compared patch sampling strategies on noisy root images with condensation. This study compares three strategies — random (baseline), weighted (H1), and hard-example (H2) — to determine which produces the best U-Net segmentation performance.

“Does patch sampling strategy affect U-Net segmentation performance on plant root images under condensation and noise?”

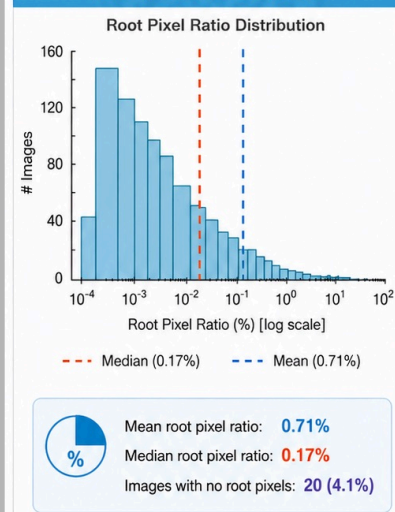
## DATASET

Dataset collected by the Hades in-vitro phenotyping system at NPEC (Netherlands Plant Eco-phenotyping Centre). Images are greyscale morphometric scans of Arabidopsis thaliana root systems grown in petri dishes. Each dish was sown with 5 seeds and photographed daily. Visual variation arises from condensation forming on the petri dish surface.

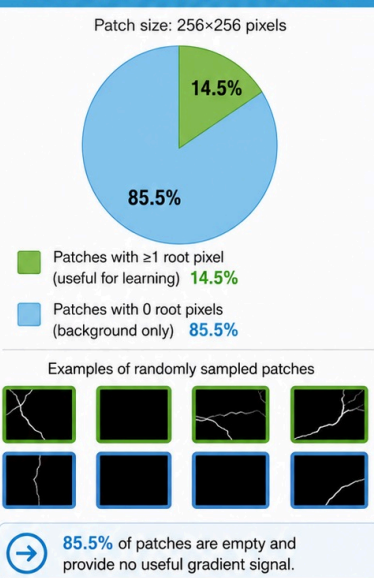
| Cohort | Total Images | Train (70%) | Validation (15%) | Test (15%) |
|--------|--------------|-------------|------------------|------------|
| Y2B_23 | 115          | 81          | 17               | 17         |
| Y2B_24 | 314          | 220         | 47               | 47         |
| Y2B_25 | 63           | 44          | 10               | 9          |
| Total  | 492          | 344         | 74               | 74         |

## EDA INSIGHTS

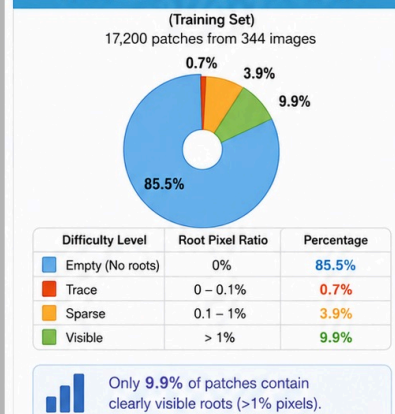
### 1. CLASS IMBALANCE



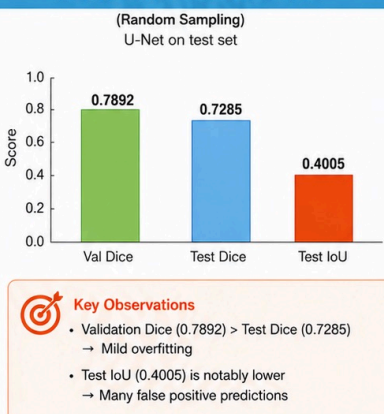
### 2. RANDOM PATCH SAMPLING



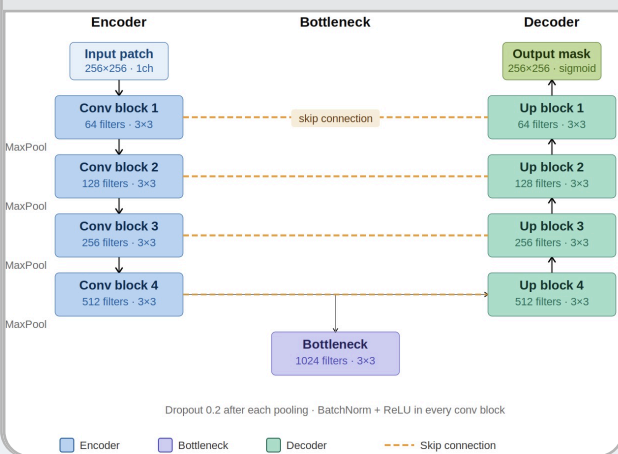
### 3. PATCH DIFFICULTY ANALYSIS



### 4. BASELINE PERFORMANCE



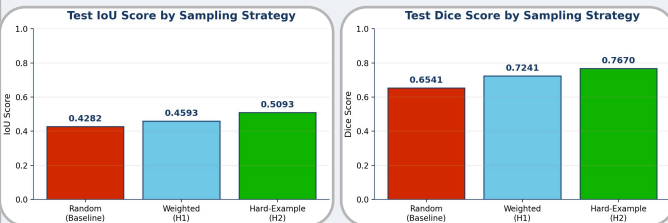
## U-NET ARCHITECTURE



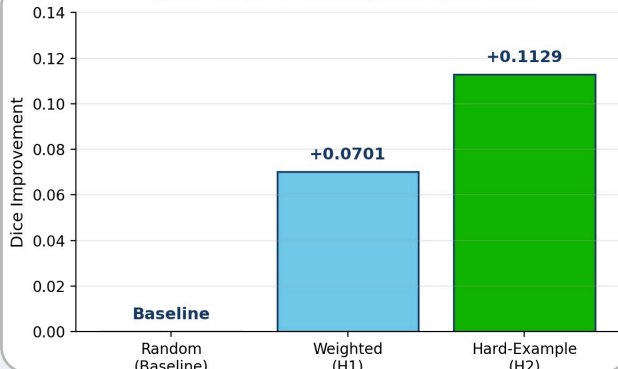
## U-NET ARCHITECTURE

| Model used for all three patch sampling strategies |                                       |
|----------------------------------------------------|---------------------------------------|
| Component                                          | Details                               |
| Input                                              | 256x256 greyscale (1 channel)         |
| Encoder blocks                                     | 4 downsampling blocks                 |
| Conv layers/block                                  | Two 3x3 convolutions                  |
| Normalisation                                      | Batch normalisation + ReLU            |
| Pooling                                            | 2x2 max pooling                       |
| Feature channels                                   | 64 → 128 → 256 → 512                  |
| Bottleneck                                         | 1024 filters                          |
| Decoder                                            | 4 upsampling blocks + transposed conv |
| Skip connections                                   | Encoder ↔ Decoder at each level       |
| Final layer                                        | 1x1 convolution + sigmoid             |
| Dropout                                            | 0.2 after each pooling layer          |
| Total parameters                                   | 31,042,369                            |
| Framework                                          | PyTorch 2.12                          |
| GPU                                                | NVIDIA RTX 6000 Ada Generation        |

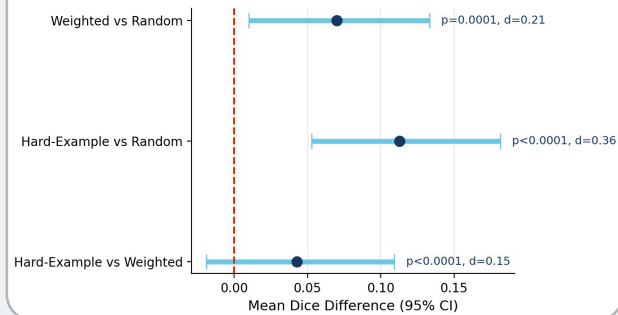
The architecture used across all 3 conditions — differences in results attributable solely to patch sampling strategy



### Dice Improvement Over Baseline



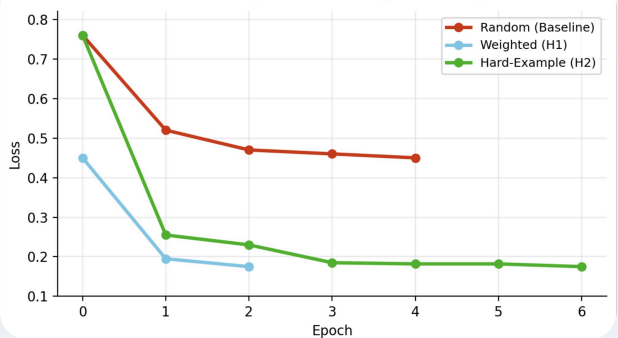
### Statistical Comparison of Dice Improvements



### Validation Dice During Training



### Training Loss During Training



## HYPOTHESIS EVALUATION

| Random                                                                             | Weighted                  | Hard-Example              |
|------------------------------------------------------------------------------------|---------------------------|---------------------------|
| Dice 0.6541<br>IoU 0.4282                                                          | Dice 0.7241<br>IoU 0.4593 | Dice 0.7670<br>IoU 0.5093 |
| Hard-Example achieved the best overall result<br>+0.1129 Dice over Random baseline |                           |                           |

### SUPPORTED

**H1 Supported** — Weighted sampling achieved Dice 0.7241 vs baseline 0.6541 (+0.0701, p=0.0001, d=0.21). Concentrating patches on root-containing regions directly reduces class imbalance and improves segmentation.

### SUPPORTED

**H2 Supported** — Hard-example sampling achieved the highest Dice of 0.7670 (+0.1129 vs baseline, p<0.001, d=0.36). Forcing training on difficult condensation-affected patches yields the strongest performance.

Key finding  
sampling strategy matters

## CONCLUSIONS

Patch sampling strategy significantly affects U-Net segmentation performance on plant root images under condensation and background noise. Both proposed strategies outperformed the random baseline, with hard-example sampling achieving the highest Dice of 0.767 — a 17% improvement. These gains were achieved without any changes to the model architecture, demonstrating that patch selection is a simple but impactful design decision in segmentation tasks with severe class imbalance. The advantage of hard-example over weighted sampling was statistically significant but variable across patches, highlighting that both strategies have value depending on the imaging conditions.

## LIMITATIONS

- Small test set (74 images) — limited statistical power
- Only one training run per strategy — results may vary across runs.
- Single U-Net architecture — may not generalise to attention U-Net or nnU-Net
- Single dataset domain — one lab, one plant species
- Error maps updated every 10 epochs instead of 5 due to GPU constraints

## FUTURE WORK

- Repeat each experiment 3–5× to measure training variability
- Test hybrid strategy: weighted + hard-example combined
- Evaluate on attention U-Net, nnU-Net, TransUNet
- Validate on larger multi-species datasets
- Fully online OHEM with per-batch error map updates
- Full-image evaluation instead of patch-level only

## REFERENCES

Ronneberger et al. (2015)  
Isensee et al. (2018)  
Khan et al. (2020)  
Sharma et al. (2021)  
Santurkar et al. (2019)